

Exoplanet Discovery Using Pulsar Timing



Henry B, Sam H, Sahil S and Will T

Contents Page

1. Scientific Pulsar Background

- 1.1. What are pulsars?
- 1.2. Pulsar timing
 - 1.2.1. Method of pulsar timing
 - 1.2.2. Use for Exoplanet Discovery
- 1.3. PSR 1719 - 1438

2. Preliminary Results

- 2.1. Introduction to Libstempo
- 2.2. Deriving Parameters about the pulsar
- 2.3. Proving the existence of the planet
 - 2.3.1. Phase folding & Wrap around function

3. Method

- 3.1. Discovering the Period
- 3.2. Initial Calculations Using the period, P
 - 3.2.1. Orbital radius
 - 3.2.2. Mass of the companion
- 3.3. Finding a more accurate period

4. Conclusion

- 4.1. Future ambitions

What are Pulsars?

How are pulsars formed?

- Formed after collapse of a massive star.
- Remaining stellar object is an extremely dense core - neutron star.
- During collapse some neutron stars can begin to spin due to magnetic fields.



Arecibo 305m Telescope
(collapsed 2020) -
<https://www.astronomy.com/science/battered-but-not-broken-arecibo-observatory-receives-renewed-funding/>

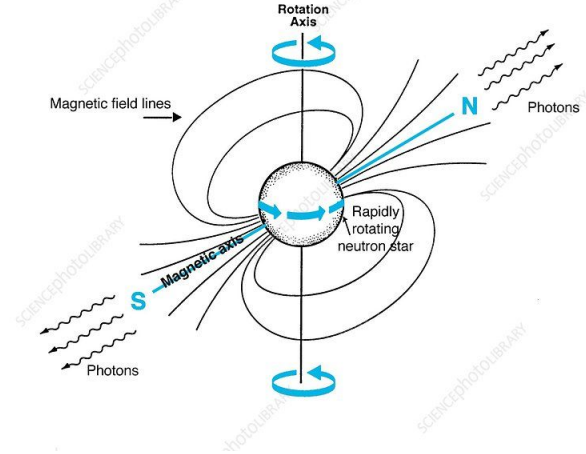


Diagram briefly showing how a pulsar work -
<https://www.sciencephoto.com/media/331715/view/pulsar>

Characteristics of pulsars

- Neutron star rotating at extreme speeds.
- Extremely strong magnetic fields.
- Accelerates particles to relativistic speeds.
- Releases jets of lights at along poles.
- Pulses monitored using radio telescopes.

Pulsar Timing

How does it work?

- Radio telescopes detect regular interval of pulses
- Period = time between emitted pulses.
- Typically this is fairly stable.

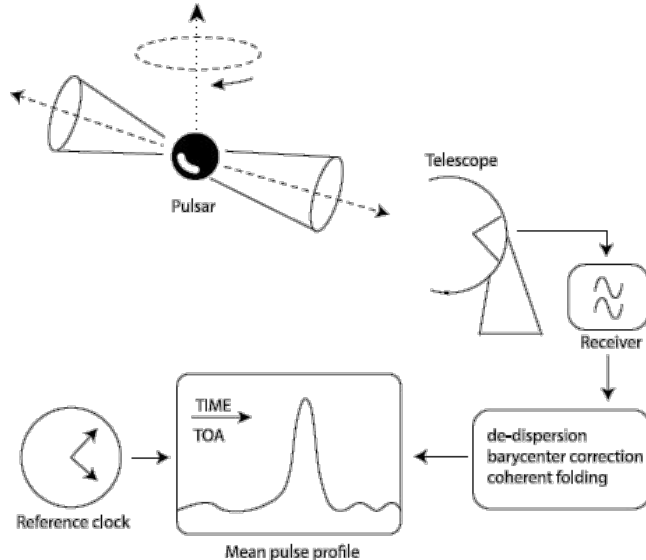


Diagram showing how pulsar timing allows us to use pulsars as clocks - <https://link.springer.com/article/10.1007/s11214-017-0459-0>

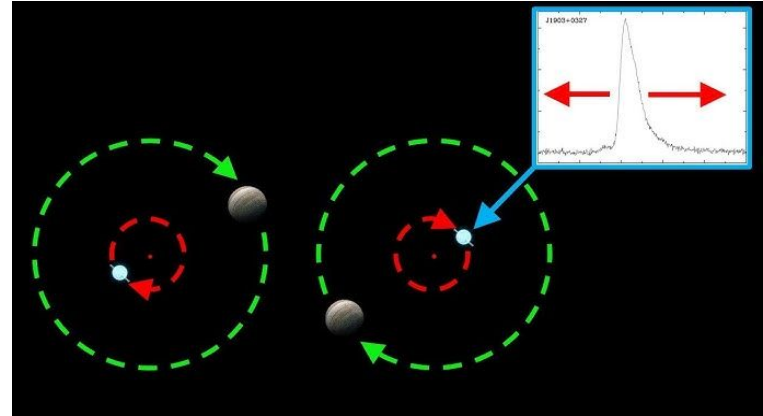


Image showing how pulsar timing can detect period irregularities - <https://www.youtube.com/@AstroPhil2000>

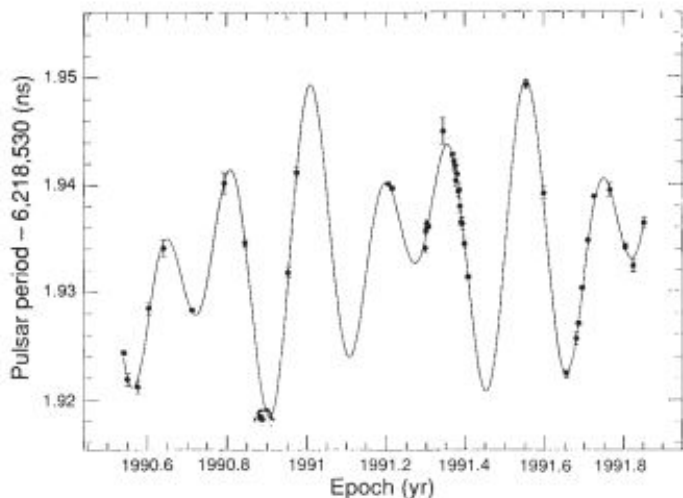
How can we relate this to exoplanet detection?

- Sometimes, small variations in period can be detected.
- Must be in a binary with a planet.
- Irregularities can be examined in more detail.

Pulsar Timing

History of pulsar timing

- First ever exoplanets detected - Wolszczan & Frail 1992.
- In orbit around PSR 1257+12.
- Used the 305m Arecibo telescope.
- 0.7m/s 'wobble' derived.



Period differences of PSR 1257+12 derived from post-fit - Wolszczan, A., Frail, D., 1992

Planet Name	Host Name	Default Parameter Set	Number of Stars	Number of Planets
☒ M62H b	M62H	1	1	1
☒ PSR B0329+54 b	PSR B0329+54	1	1	1
☒ PSR B1257+12 b	PSR B1257+12	0	1	3
☒ PSR B1257+12 c	PSR B1257+12	1	1	3
☒ PSR B1257+12 c	PSR B1257+12	1	1	3
☒ PSR B1257+12 c	PSR B1257+12	0	1	3
☒ PSR B1257+12 d	PSR B1257+12	0	1	3
☒ PSR B1257+12 d	PSR B1257+12	0	1	3
☒ PSR B1257+12 d	PSR B1257+12	1	1	3
☒ PSR B1620-26 b	PSR B1620-26	1	2	1
☒ PSR J1719-1438 b	PSR J1719-1438	1	1	1
☒ PSR J2322-2650 b	PSR J2322-2650	1	1	1

Table showing all the exoplanets on NASA's database detected via pulsar timing - <https://exoplanetarchive.ipac.caltech.edu/>

How has pulsar timing been used to detect exoplanets since?

- 8 confirmed exoplanets on NASA's database
- A few interesting systems to highlight.
- PSR 1257+12 has 3 exoplanets in orbit.
- PSR 1620-26 is in an orbital system with a star and planet.

Aims of the Experiment

Starting point:

- Interesting how few exoplanets discovered via pulsar timings.
- Is there a reason for this? Is it to do with the method?
- Decided we would try and recreate a discovery, test the difficulty of it.

Finding a planet:

- Wanted a relatively simple system with data easily available
- Aim to find as many details about an exoplanet as possible (e.g mass, orbital radius etc.)

The Pulsar System

The Pulsar:

- The pulsar we are studying is PSR J1719-1438
- Found in 2010 as one of 5 millisecond Pulsars found in The High Time Resolution Pulsar survey
- The pulses of radio waves it emits have a period 5.7ms
- Measurements were taken with the Lovell 76-m telescope in Cheshire and the Parkes 64m telescope in New South Wales

Why we chose it:

- Binary system with a single Pulsar and single planet
- Data is easily accessible through (Bailes et. al, 2011), and we can compare values as we go
- Therefore a good planet to practice the method on



64m Parkes radio telescope
(https://en.wikipedia.org/wiki/Parkes_Observatory)

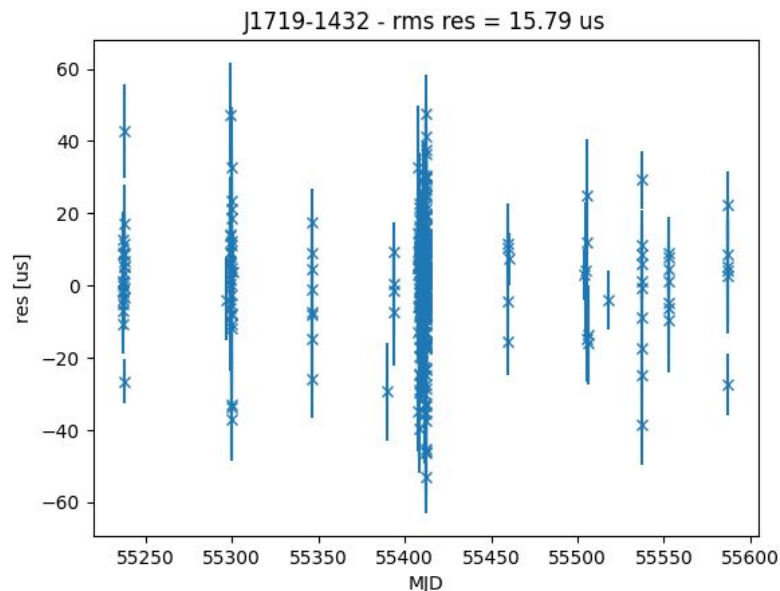
Introduction to Libstempo

- A python wrapper for Tempo2
- .par and .tim files used to construct residual graph

Parameter	Value	Mode 0 or 1	Error on value
PSRJ	J1719-1432		
RAJ	17:19:10.0729928	1	0.00006943603747332227
DECJ	-14:38:00.96484	1	0.00944194834215492708
F0	172.70704460192937908	1	0.0000000019088425374
F1	-2.1958832200264405067e-15	1	1.3410010308008839596e-17
PEPOCH	55235.51740116860422		
POSEPOCH	55157.771186275891957		
DMEPOCH	55235.517213687242474		
DM	36.766		
BINARY	ELL1		
EPS1	0.0		
EPS2	0.0		
TASC	55235.516524392188234	1	
PB	0.090706291234533751382	1	0.00000000868810462674
T0	55235.516524392188234	1	0.00001668746422891375

1368.87500000	55412.41105531144520108	8.84000	7
1368.87500000	55412.41140047914710109	6.04000	7
1368.87500000	55412.41174771360289952	8.25000	7
1368.87500000	55412.41209494790910028	10.56000	7
1368.87500000	55412.41244218242000130	6.89000	7
1368.87500000	55412.41278941722410067	9.23000	7
1368.87500000	55412.41313658454309987	9.78000	7
1368.87500000	55412.41348381945200074	9.24000	7
1368.87500000	55412.41383105341930104	9.72000	7

Frequency of observation MHz	Time pulse arrived in MJD + fraction of the day	TOA Uncertainty	Telescope ID (Parkes telescope)
------------------------------	---	-----------------	---------------------------------



Residual against MJD* Graph of actual data about PSR 1719-1438, gaps show the missing data entry points. * MJD refers to the day after midnight of November 17th, 1858

Derived Parameters about the Pulsar

Period of the Pulsar:

- $P = 1/f$
- $dP/dt = -1/f^2 df/dt \therefore \dot{P} = -\dot{f}/f$

$P = 0.005790151770038619 \text{ s}$ $\dot{P} = 7.361886796534389\text{e-}21 \text{ s/s}$

Surface Magnetic Field:

- $B_s = 3.2 \times 10^{19} \sqrt{\dot{P}P} \text{ G} \approx 10^{21} (\dot{P}/10^{-15})^{0.5} (P/\text{s})^{0.5} \text{ G} \approx 206461720.09788528 \text{ G} \text{ (} 2.1 \times 10^8 \text{ G)}$

Characteristic age:

- $\tau = P/2\dot{P} \approx 3.932518884129226\text{e+}17 \text{ yrs} \approx 4\text{e+}6 \text{ Gyrs}$

PSRJ	J1719-1432		
RAJ	17:19:10.0729928	1	0.00006943603747332227
DECJ	-14:38:00.96484	1	0.00944194834215492708
F0	172.70704460192937908	1	0.00000000019088425374
F1	-2.1958832200264405067e-16	1	1.3410010308008839596e-17
PEPOCH	55235.51740116860422		
POSEPOCH	55157.771186275891957		
DMEPOCH	55235.517213687242474		
DM	36.766		

Parameters of PSR 1719-1438 from the .par file showing the frequency and frequency derivative with errors. [M. Bailes et al.

,Transformation of a Star into a Planet in a Millisecond Pulsar

Binary.Science333,1717-1720(2011).DOI:[10.1126/science.1208890](https://doi.org/10.1126/science.1208890)]

Derived parameters	
Characteristic age (Gy)	>12.5
B (G)	<2 × 10 ⁸

Derived parameters given by M. Bailes in the paper [M. Bailes et al.

,Transformation of a Star into a Planet in a Millisecond Pulsar

Binary.Science333,1717-1720(2011).DOI:[10.1126/science.1208890](https://doi.org/10.1126/science.1208890)]

Derived Parameters about the Pulsar

Spin-down luminosity - essentially the lost rate of rotational energy

$$-\dot{E} = 4\pi^2 I \dot{P} P^{-3}$$

Where I is the moment of inertia

$$I = 2MR^2 / 5 \rightarrow \rho = 3M / (4\pi R^3) \approx 10^{45} \text{ g cm}^3$$

$$-\dot{E} = 1.497197279338788 \times 10^{33} \text{ erg/s}$$

$$-\dot{E} = 0.389286864102649 \approx 0.40 L_{\odot}$$

sun

Derived parameters	
Characteristic Age (Gyr)	>12.5
B (G)	<2×10 ⁸
Dispersion Measure Distance (kpc)	1.2 (3)
Spin-down Luminosity $L_{\odot}$	<0.40(4)

Derived from a single observation

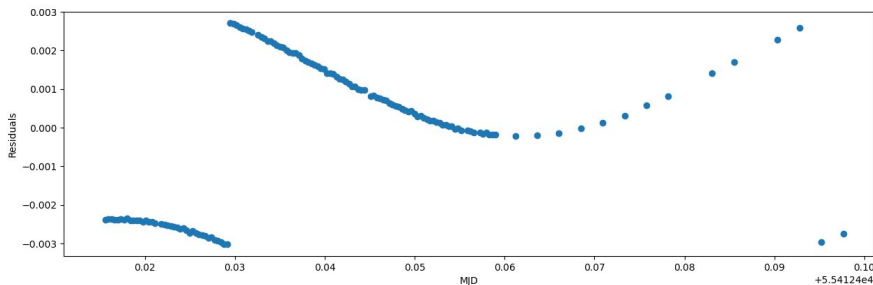
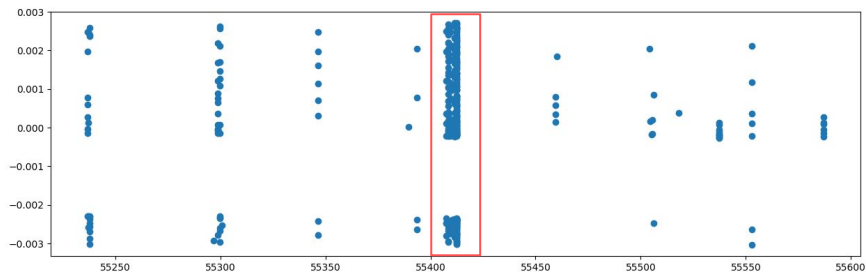
Derived Parameters from bailes' paper

Where, $L_{\odot}$ is the luminosity of the

Proving the existence of the Planet

- Used matplotlib to be able to manipulate data better
- Removed binary parameters from .par file for libstempo
- Zoomed into MJD 55412 shows one period

Residual against time, for pulsar without binary parameters



```
BINARY          ELL1
EPS1 0.0
EPS2 0.0
TASC 55235.516524392188234      1
PB          0.090706291234533751382      1      0.00000000868810462674
T0          55235.516524392188234      1      0.00001668746422891375
A1          0.0018197858165632653794      1      0.00000076745583613271
START       55236.945585683606623      1
FINISH      55586.995444416301325      1
TZRMJD      55412.411053311453024
TZRFREQ     1368.875
TZRSITE      7
TRES        14.541
EPHVER       5
CLK          TT(TAI)
MODE 1
EPHEM        DE200
NITS         1
NTOA         388
MODE 1
```

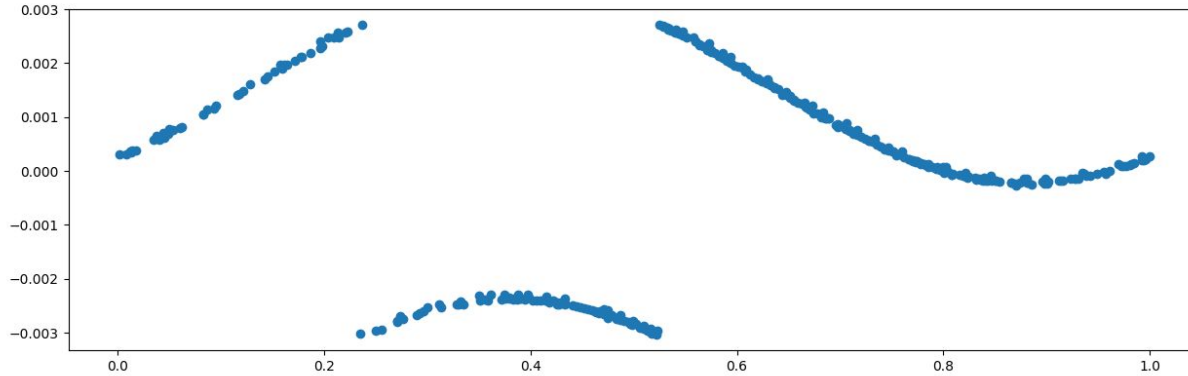
Binary parameters for PSR 1719-1438 such as the companions' period, semi-major axis orbit.

Applied two functions: Fold and Wrap around

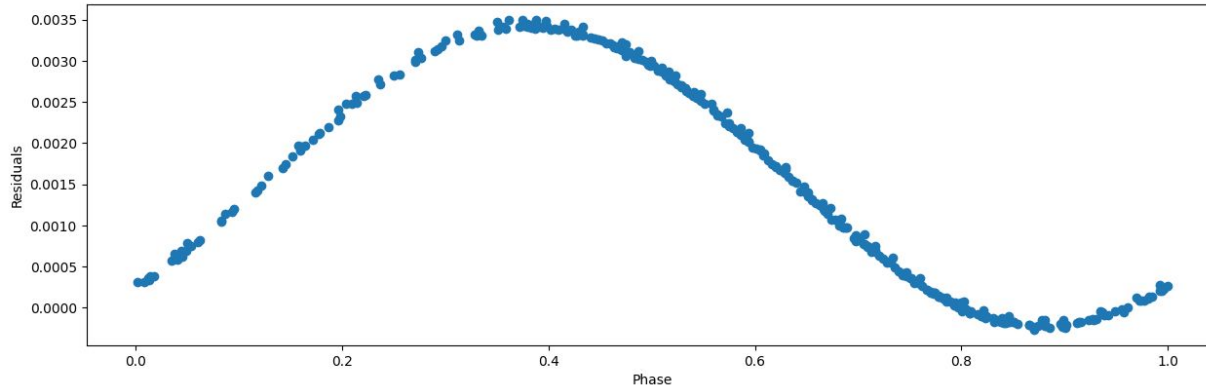
Fold function: $((\text{time} - t_0)/P_b) \% 1$

Wrap around Function: If residual < 0.02, residual + period else residual

Residuals against phase, with both fold function and wrap around function



Only fold function to all data points

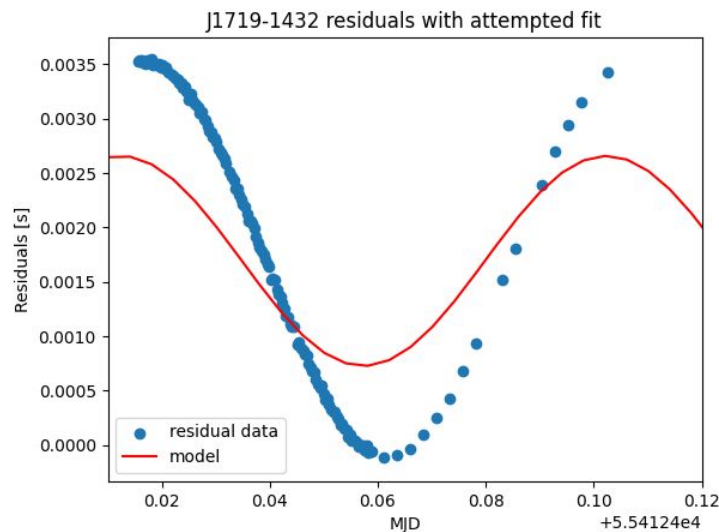
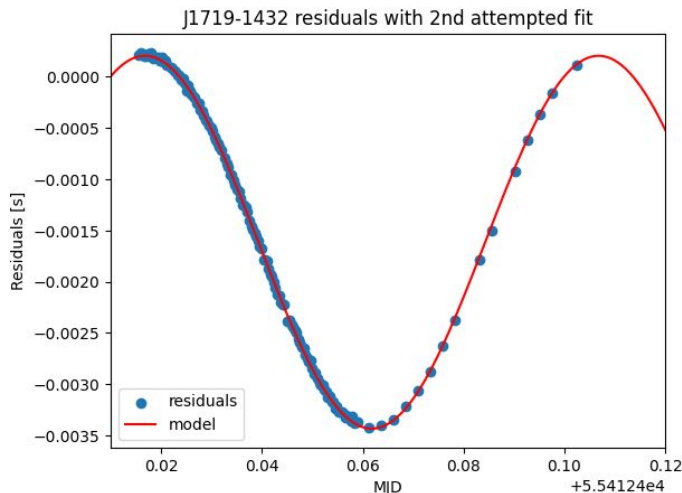


Both fold function and wrap around function applied showing one complete phase

Period discovery

Initial attempt:

- After we plotted residuals, set about finding the binary period.
- Predominately used `scipy.optimize` curve fit to attempt to fit a sine wave
- Initially tried fitting sine wave to all the data at once, but failed to work.



Second try:

- Limited the data to a single day
- Inaccurate as does not use the whole data set
- Gave initial value of about 0.091 days

Calculations using initial period P

Orbital Radius of the Planet:

$$T^2 = (4\pi^2 / Gm_p) R^3$$

P = Period [days] = 0.09099998539977734

T = Orbital period [s] = **P** × 24 × 3600

m_p = Pulsar mass [kg] = 1.4M_⊙

G = Gravitational Constant [Nm²kg⁻²] = 6.67430 × 10⁻¹¹

R = Orbital Radius [m]

Planet Radius: Unknown	Planet Type: Gas Giant	Discovery Method: Pulsar Timing	Planet Mass: 1.2 Jupiters
Discovery Date: 2011	Orbital Radius: 0.0044 AU	Orbital Period: 0.1 days	Eccentricity: < 0.06

Table from NASA's database showing properties of PSR J1719-1438 b - <https://science.nasa.gov/>

Substituting and rearranging:

$$R = 6.63 \times 10^8 \text{ m (3.s.f)} = 0.00443 \text{ au}$$

Agreed with values on both Bailes 2011 and NASA's database

Calculations using initial period P

$$\text{[1]} f(m_c) = (4\pi^2/G) (a_p \sin[i])^3 / T^2 \quad \text{[2]} f(m_c) = (m_c \sin[i])^3 / (m_c + m_p)^2$$

$a_p \sin[i] = 0.001819$ light seconds - convert to m

m_c = Companion mass [kg]

T and m_p as previously defined

- Assume an edge on orbit; $\sin[i] = 1$
- Assume $m_c \ll m_p$

Result:

Via [1], $f(m_c) = 7.801 \times 10^{-10} M_\odot$

Substitute into [2] and take assumptions; $m_c^3 / m_p^2 = 7.801 \times 10^{-10} M_\odot$

Results in $m_c = 2.291 \times 10^{27} \text{ kg} = 1.207 M_J$ (M_J = Mass of Jupiter)



Image of Jupiter, a very comparable planet to PSR J1719-1438 b - <https://en.wikiversity.org/wiki/Jupiter>

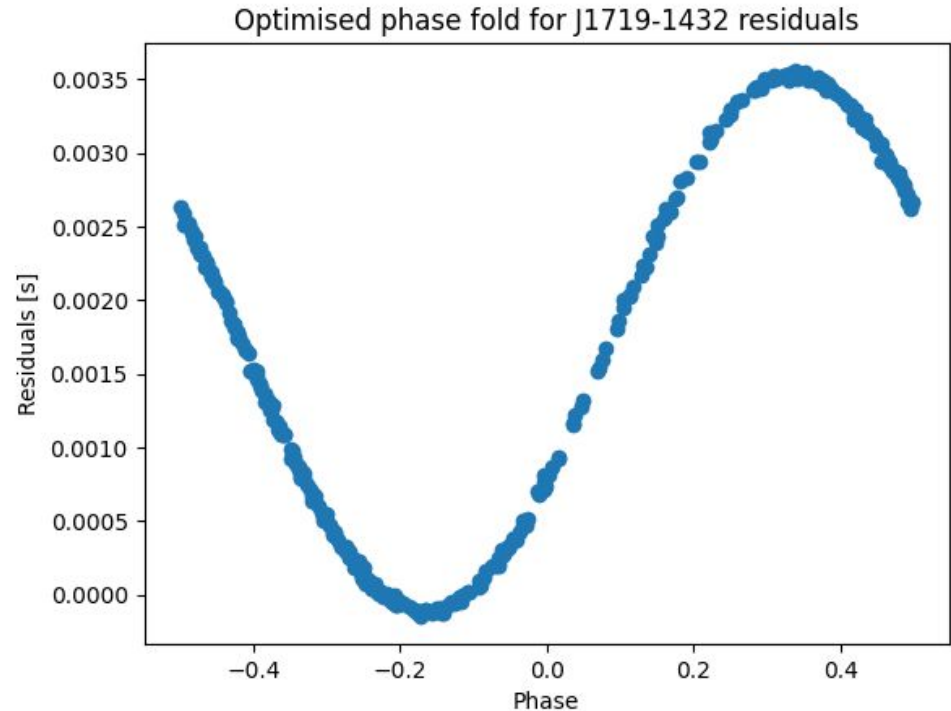
Finding a more accurate period

Phase folding:

- Phase folded residuals using estimated period
- Wrote a function to fit a sine wave with a period of 1 for a given phase fold
- Least squares optimised for a fold period

Results:

- 0.090706
- Yet to properly calculate errors



Future ambitions

- Calculate the radius of the planet
- An upper estimate for the radius can come from the Roche Lobe radius. Which is approximated using:

$$R_L = 0.462a (m_c / (m_c + m_p))^{1/3}$$

Where:

- R_L = Roche Lobe radius
- a = the combined separation of the two bodies
- Under the assumption $m_c \ll m_p$, Kepler's law gives a good estimate for a :

$$a = 0.95R_\odot (m_p / 1.4 M_\odot)^{1/3}$$

- Calculate the Density of the planet

- A lower estimate for the density of the planet can come from the mean density-orbital relation

$$\rho = 3\pi / (0.462^3 GP_b)$$

- It is derived from Roche Lobe radius and the mass function

Conclusion

- Briefly outlined **what pulsars are** and analysed the **method of pulsar timing**.
- Presented the **aims** of our experiment and how we initially planned to achieve these.
- Outlined **PSR J1719-1438 and Bailes' 2011 paper** and how these are relevant to our research.
- Introduced **libstempo** and how this can be used to create figures from which we calculated the **period of the planet** in orbit.
- Using libstempo derived several **key parameters** explaining the binary system.
- Did several **calculations** from our derived period to obtain values for both orbital radius **R** and also the mass of the companion **m_c** .
- Resolved a **more accurate period** using phase folding.
- Concluded our **plans moving forward** for the rest of the research period and how we aim to go about these.